

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title

Milijuli – Group Savings & Loan Management System (PWA)

1. INTRODUCTION

1.1 Purpose

This document defines the requirements for a web-based **Group Savings & Loan Management System** for a real-world group/sahakari environment where financial records must be transparent to all members.

The system allows:

- Two admins to manage financial records
- Members to view all group transactions
- Only admins can modify records
- Private authentication data remains hidden

1.2 Scope

The system is a Progressive Web Application (PWA) where:

- Users register and wait for admin approval
- Two admins manage all financial records
- All members can see group financial data
- Members cannot edit data
- Authentication data is private
- Transactions can include optional photo proof

The system replaces manual bookkeeping methods such as paper records or Excel sheets.

1.3 Definitions

Admin: System manager with full financial control

Member: Group participant with read-only access

Transaction: Savings, Loan, or Payment record

PWA: Web app that behaves like a mobile application

Sahakari: Group/community-based savings and loan environment

2. OVERALL DESCRIPTION

2.1 System Overview

Web-based application (PWA)

Roles:

Admin (2 users)

- Equal authority
- Can manage all data
- Can approve users
- Can add/edit/delete transactions
- Can manage reports

Member

- Can log in
- Can view all group transaction records
- Can view summaries and balances
- Cannot modify anything

2.2 User Classes

Admin

Two admins exist in the system.

Both can:

- Manage users
- Manage savings
- Manage loans
- Manage payments
- View reports
- Edit/delete records
- Upload proofs

No admin hierarchy (both equal).

Member

Members can:

- View all financial transaction records

- View group reports
- View balances
- Search transactions

Members cannot:

- Add records
- Edit records
- Delete records
- Access private authentication information

2.3 Operating Environment

- Modern browsers (Chrome, Firefox, Edge)
- Mobile support
- Desktop support
- Internet required

2.4 Assumptions

- Admins are responsible for financial correctness
- Members use the system for transparency
- Data visibility improves trust
- Members cannot modify records

3. SYSTEM FEATURES

3.1 Authentication System

Registration

Users register with:

- Full name
- Phone number
- Email
- Password

Status:

- Pending
- Approved

- Approved
- Rejected

Login

Only approved users can log in.

Role-based login:

- Admin
- Member

3.2 User Profile

Each user contains:

- Member ID
- Full Name
- Phone Number (private)
- Address
- Email (private)
- Password (encrypted/private)
- Profile Photo
- Joined Date
- Account Status
- Role

3.3 Privacy Rules

Visible to All Members

- Member Name
- Transaction history
- Savings records
- Loan records
- Payment records
- Reports
- Financial summaries
- Receipt numbers

- Notes
- Dates

Hidden from Members

- Phone numbers
- Emails
- Passwords
- Login credentials
- Account security details

Only admins can access private member details.

3.4 Savings Module

Admin

Can:

- Add savings entry
- Edit savings entry
- Delete savings entry
- Upload proof photo
- Assign transaction ID

Member

Can:

- View all savings records
- View totals
- Search records

Cannot edit.

3.5 Loan Module

Admin

Can:

- Create loan
- Edit loan

- Delete loan
- Upload proof
- Assign transaction ID

Member

Can:

- View all loan records
- View loan history
- View balances

Cannot edit.

3.6 Payment Module

Admin

Can:

- Add payment
- Link payment to loan
- Edit payment
- Delete payment
- Upload proof

Member

Can:

- View all payment records
- View repayment history

Cannot edit.

3.7 Transaction System

Each transaction includes:

- Transaction ID / Receipt Number

- Member Name
- Amount
- Date
- Type
- Note
- Optional Photo Proof
- Created by Admin

All members can see transaction records.

3.8 Reports Module

Admin can view:

- Total group savings
- Total loans issued
- Total payments received
- Member-wise summaries
- Monthly reports
- Search/filter reports

Members can also view:

- Same financial reports (read-only)

3.9 Search & Filters

Members and Admins can search by:

- Member name
- Member ID
- Transaction ID
- Date
- Transaction type

Admins have edit rights; members only view.

4. EXTERNAL INTERFACES

4.1 User Interface

Admin Dashboard

- User management
- Transaction entry
- Reports
- Edit/delete controls

Member Dashboard

- View-only dashboard
- Transaction history
- Reports
- Search/filter

4.2 Software Interfaces

- Cloud database (Supabase recommended)
- Cloud storage for proofs/images
- Authentication system

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Security

- Encrypted passwords
- Role-based access
- Private auth data hidden from members
- Admin-only write access

5.2 Transparency

- All members can view financial records
- No hidden transaction data
- Shared visibility improves fairness

5.3 Performance

- Fast loading (<3 sec)
- Optimized queries

5.4 Reliability

- No unauthorized modification
- Data consistency maintained

5.5 Scalability

- Support many members
- Support increasing transactions

6. DATA REQUIREMENTS

Core entities:

- Users
- Savings
- Loans
- Payments
- Transactions
- Reports

7. VERSION 1 EXCLUSIONS

Not included in V1:

- Automation
- SMS notifications
- Email notifications
- Penalty system
- Auto-interest calculation
- Payment gateway

8. CONSTRAINTS

- Internet connection required
- Admin responsible for financial correctness
- Members are view-only
- Two admins have equal power

